positives. The present network meta-analysis[1] does not adequately address these methodological hazards, impairing its ability to detect only true treatment differences.

In the context of network meta-analysis, transitivity is the crucial assumption that studies share similar clinical and design characteristics relevant to estimating an effect size. Transitivity permits the use of indirect evidence—that is, it permits the comparison of treatments that have never been directly contrasted. Without transitivity, any indirect evidence might be misestimated, since different treatments might have been tested in different contexts, such as varying degrees of disease severity. Cipriani and colleagues[2] provide an example of violation of transitivity with the treatments A, B, and C:

"Suppose that all AC studies include patients with severe illness and all BC studies include patients with moderate illness. Each study set is similar within itself… but the two sets deal with clinically different populations of patients. So, if severity is an effect modifier, the transitivity assumption would not hold, and synthesis of these two meta-analyses would not give a valid AB estimate."[2]

To summarise, indirect AB comparisons will be inaccurate because treatment A tends to be tested among more intractable patients than does B.

In this network meta-analysis,[1] transitivity per se is left unmentioned, and no moderator analyses were done. As study-characteristic heterogeneity is the norm in psychiatric trials, transitivity cannot be assumed. These omissions are therefore problematic. Moreover, some treatments contributed few studies. Treatments contributing fewer studies might not portray a representative range of treatment contexts, biasing an effect estimate compared with those with more studies.

Unsurprisingly, the authors detect significant effect heterogeneity.[1] Furthermore, at least nine of 44 comparisons with both direct and indirect evidence were potentially inconsistent,[1] meaning that direct evidence was in significant disagreement with indirect evidence. In such a case, treatment A could be equivalent in direct comparison with treatment B, while indirect evidence about treatment B used in other trials estimates that treatment B is better than treatment A. It would have been judicious to highlight the disputed comparisons indicated by inconsistency. Importantly, both inconsistency and heterogeneity can imply violation of transitivity.

Simulations suggest that the risk of false positives is high in network meta-analysis because of the sheer number of comparisons made.[3,4] In a simulation comparing only 12 antidepressants, an average of 2·7 false positives were recorded.[3] That 38 active treatments and three controls were compared in this network meta-analysis[1] is concerning. Type-I error correction (eg, Bonferroni) would have been appropriate to use.

It is unfortunate that these significant limitations are largely unaddressed, since they set the stage for the detection of false superiorities between treatments. Until these concerns are resolved, it is inappropriate to make stark treatment recommendations based on indirect evidence provided by this network meta-analysis.

I have previously published a standard meta-analysis on the effects of psychodynamic therapies for anxiety disorders, including social anxiety disorder.

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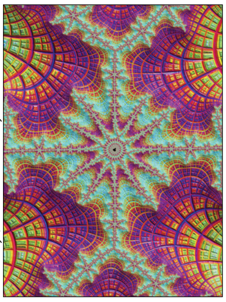
Author's reply

John Keefe criticises our network meta-analysis[1] of treatments for social anxiety for not addressing the issue of "transitivity". The introduction of the term transitivity[2] into discussions of network meta-analysis has created great confusion. We do not use the term, and instead follow the more usual practice of using the term "inconsistency",[3,4] a topic that is fully addressed in our Article.[1]

Keefe cites our comment that "there was potential for inconsistency in nine of the 44 loops in the network"[1] as meaning that "direct evidence was in significant disagreement with indirect evidence". In fact, we were simply pointing out that there were nine evidence loops for which inconsistency between direct and indirect evidence could be compared. In the event, there was no evidence of disagreement. We make this clear in the Article by stating that we found "no substantial differences in magnitude and direction between the results of the network meta-analysis and the results of the pairwise comparisons".[1]

As Keefe notes, there was evidence of statistical heterogeneity. This heterogeneity is common in both pairwise and network meta-analyses. However, the level of between-trials variation (median SD 0·19) was surprisingly small in relation to the effects of the treatment classes, most of which were between 0·8 and 1·2 SD units. This finding, again, points to a low risk of inconsistency. It is also consistent with our baseline comparisons between trials of different treatment types. As stated in the Article, "we did not identify any systematic differences in participant demographics or initial symptom severity".[1] The latter covered the presence/absence of avoidant personality disorder (when reported), as well as overall severity of social anxiety.

Keefe urges the use of Bonferonni statistics to control for the risk of false positive identification of the best treatment when many comparisons are being made, and cites a study[5]



Mehau Kulyk/ Science Photo Library

reporting bias in treatment rankings. However, network meta-analysis does not make multiple comparisons that can be corrected in this way, and the study concerned[5] does not in fact demonstrate any bias, but only shows that the posterior distribution of rankings is sensitive to the pattern of data–hardly a surprising result.

In any case, conclusions on which treatment is best are never based on the posterior distribution of the ranks, which are notoriously unstable, but on the posterior expected effect size, which is what is reported in our study. Decisions based on imperfect evidence can always be wrong, but network meta-analysis is a method that helps identify the optimum decision in view of the evidence.

DMC is the developer of one of the versions of individual CBT that was shown to be efficacious. All other authors declare no competing interests.

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Protecting the human rights of people who use psychedelics

In a recent Comment Ben Sessa[1] explained how the War on the Drugs worldwide has impeded development of psychiatric treatment with psychedelics such as LSD (lysergic acid diethylamide) and psilocybin (found in magic mushrooms). Prohibition also had negative outcomes for the millions of individuals who find it worthwhile to use psychedelics in various cultural settings outside of those in the clinic.

People have used psychedelics in spiritual practice for at least 5700 years, pre-dating all major organised religions.[2] 100 years ago, members of rival religious groups campaigned against Native American use of psychedelic peyote cactus.[2] However in the 1950s, concerned scientists used evidence and human rights arguments to defend peyote users, leading to legal exemptions for specific groups.[2] When psychedelics spread to the wider society in the 1960s, this was also acknowledged by religious scholars and governments as a spiritual movement (eg, the UK Home Office[3]).

Under the UN 1971 Convention on Psychotropic Substances, WHO has responsibility to evaluate international policy for psychedelic substances. The original WHO assessment said that psychedelics "are usually taken in the hope of inducing a mystical experience leading to a greater understanding of the users' personal problems and of the universe".[4] The WHO assessment did not cite a single example of harm from naturally-occurring psychedelics such as psilocybin or peyote, and cited only a handful of anecdotes related to LSD.[4] This was in no way an evidence-based harm assessment.

In the past 50 years, people are thought to have used at least half a billion doses of psychedelic drugs. Psilocybin mushrooms and other psychedelics are legally sold in The Netherlands. Based on extensive human experience, it is generally acknowledged that psychedelics do not elicit addiction or compulsive use and that there is little evidence for an association between psychedelic use and birth defects, chromosome damage, lasting mental illness, or toxic effects to the brain or other body organs.[2] Although psychedelics can induce temporary confusion and emotional turmoil, hospitalisations and serious injuries are extremely rare.[2] Overall psychedelics are not particularly dangerous when compared with other common activities.[2]

In 2016 the UN will have a special meeting in New York to set the future for international drug policy. Former UN Secretary General Kofi Annan and the Global Commission on Drug Policy say that we must "Ensure that the international conventions are interpreted and/or revised to accommodate...decriminalisation and legal regulatory policies."[5] National and international policies should respect the human rights of individuals who chose to use psychedelics as a spiritual, personal development, or cultural activity.

For more on **EmmaSofia** see www.emmasofia.org

I am board leader of EmmaSofia, a non-profit organisation based in Oslo, Norway, working to increase access to quality-controlled MDMA (4-methylenedioxymethamphetamin) and psychedelics. I have received funding from the Research Council of Norway (grant 185924).

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